VEKTOR RESEARCH SERIES · 2026

AI as Instrument: Machine Learning in Systematic Portfolio Management

Where machine learning is used in Vektor, where it is not, why those boundaries exist, and how human judgement is preserved at every gate.

For institutional investors, compliance officers, and risk committees

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ABSTRACT

The term 'AI-powered' has become a marketing claim so widely applied that it has ceased to carry information. This paper takes the opposite approach: it states precisely where machine learning is used in the Vektor platform, what it does at each point, and where it is explicitly not used. The governing principle is simple: AI in Vektor is an instrument of computation, not a maker of decisions. Every investment decision requires explicit human action. This governance framework is consistent with the AI governance principles published by financial regulators in Singapore (MAS FEAT), Hong Kong (SFC/HKMA), and the United Kingdom (FCA/PRA) — three jurisdictions that have independently converged on the same four requirements.

KEY TAKEAWAYS

— ML is used in two bounded roles: universe screening optimisation and XGBoost signal validation. Neither role makes investment decisions.

— XGBoost validates signals — it does not generate, approve, or act on them. A portfolio manager must approve the complete strategy before any position is taken.

— Human approval gates exist at every capital-at-risk stage: strategy approval, allocation execution, and cash funding are explicit human actions that cannot be automated away.

— Every model is trained on explicit historical data, produces an interpretable confidence score, and can be inspected, versioned, and rolled back. No black boxes.

— MAS FEAT (Singapore), SFC/HKMA (Hong Kong), and FCA/PRA (UK) all converge on the same four requirements: explainability, human oversight, accountability, and auditability. Vektor's framework aligns with all four across all three jurisdictions.

1. THE GOVERNING PRINCIPLE

One sentence. Every other decision follows from it.

> AI in Vektor is an instrument of computation. It does not make investment decisions, client decisions, or compliance decisions. Humans make those decisions — every time, at every gate, with full information.

This principle was the starting position of the design process, not a response to regulatory pressure. The question asked at the outset: where does machine learning outperform human analysis on a clearly defined, measurable task — and where does human judgement remain essential? The boundary between those two categories is structural. It cannot be toggled in configuration.

2. WHERE MACHINE LEARNING IS USED

Two specific, bounded applications — defined input, defined output, defined role in the workflow.

APPLICATION 1 — UNIVERSE SCREENING (RESEARCH ENVIRONMENT)

Statistical analysis in the SageMaker JupyterLab environment reduces an exchange universe of 500+ instruments to a target portfolio size, applying liquidity, sector, and dividend continuity criteria.

What it does

Ranks instruments against explicit screening parameters. Every criterion is stated: minimum average daily volume, sector membership, dividend continuity threshold. The reason any instrument is included or excluded can be stated precisely.

What it does not do

It does not select the final portfolio. The screened universe is the input to portfolio optimisation — a human analyst reviews the screening output before the optimisation runs.

APPLICATION 2 — XGBOOST SIGNAL VALIDATION (PRODUCTION INFERENCE)

The XGBoost classification model validates live trading signals before they are presented to the portfolio manager for approval.

Training data

Three years of daily OHLCV price history and historical signal outcomes.

What it does

Outputs a confidence score per signal. Signals below the configured threshold are flagged and filtered from the strategy record before PM review.

What it does not do

It does not approve strategies, execute trades, assign clients, or make any capital-affecting decision. Its output is one input to the PM review.

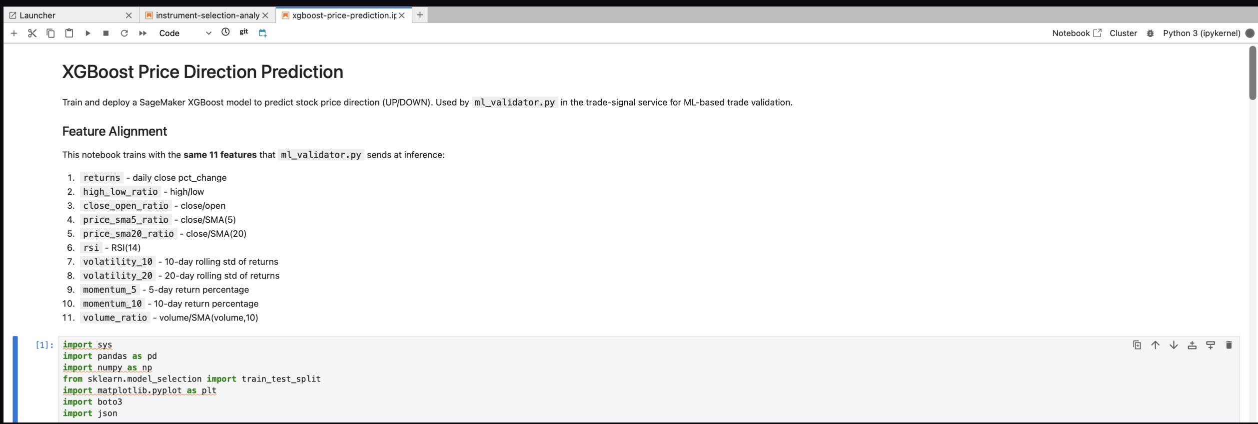
Model management

Trained, versioned, and deployed via SageMaker. Each version stored as an artefact in S3. Deployment requires an explicit promotion step — not automatic replacement. Rollback is a single operation.

Transparency

Confidence threshold is a configurable, visible parameter. Training data window and feature set are documented. Validation result — confidence score and pass/fail — stored in the strategy record and visible to PM at review.

PLATFORM EVIDENCE



XGBoost Price Direction Prediction — the 11 features used by ml_validator.py at inference. Same feature set at training and inference ensures no train-serve skew. Model: SageMaker XGBoost.

PLATFORM EVIDENCE

8. Test ML Validation Function

[10]:
```
# Test ML signal validation with real price data
test_ticker = df['symbol'].sample(1).iloc[0]
print(f"Testing with random ticker: {test_ticker}")

# Use actual price data from the selected ticker (last 30 days)
ticker_data = df[df['symbol'] == test_ticker].tail(30)
test_price_data = []
for _, row in ticker_data.iterrows():
    test_price_data.append({
        'date': row['price_date'],
        'open': row['open_price'],
        'high': row['high_price'],
        'low': row['low_price'],
        'close': row['close_price'],
        'volume': row['volume']
    })
print(f"Using {len(test_price_data)} days of actual price data")

# Test different scenarios
for signal in ['BUY', 'SELL', 'HOLD']:
    result = validate_signal_with_ml('xgboost-price-direction-endpoint', test_ticker, test_price_data, signal)
    print(f"\nML Validation for {signal} signal:")
    print(json.dumps(result, indent=2))
```

```
Testing with random ticker: BS6
Using 30 days of actual price data

ML Validation for BUY signal:
{
  "ticker": "BS6",
  "original_signal": "BUY",
  "final_signal": "WEAK_BUY",
  "ml_direction": "UP",
  "ml_probability": 0.5332627892494202,
  "confidence": "LOW",
  "agreement": "YES",
  "model_type": "SageMaker XGBoost"
}

ML Validation for SELL signal:
{
  "ticker": "BS6",
  "original_signal": "SELL",
  "final_signal": "WEAK_SELL",
  "ml_direction": "UP",
  "ml_probability": 0.5332627892494202,
  "confidence": "LOW",
  "agreement": "NO",
  "model_type": "SageMaker XGBoost"
}
```

ML Validation test output — BUY on BS6 produces WEAK_BUY (ml_direction=UP, confidence=LOW, agreement=YES). SELL produces WEAK_SELL (agreement=NO). Full JSON output stored in strategy record.

3. WHERE MACHINE LEARNING IS NOT USED

Explicit exclusions — and the reasons behind each one.

Stating where machine learning is not used is as important as stating where it is. The following exclusions are design decisions that will not change as the platform develops.

Investment decisions

No model approves, rejects, or modifies a strategy. The portfolio manager approves the complete strategy before any client assignment. This approval is a human action, recorded with timestamp and user identity.

Portfolio weight determination

Weights are produced by portfolio configuration sampling and efficient frontier optimisation via Modern Portfolio Theory — mathematical processes, not ML models. The max-Sharpe allocation is deterministic: same inputs, same output, explainable step by step.

Client suitability

Risk profiles are set and reviewed by portfolio managers. No model assesses whether a client is suitable for a given strategy.

Compliance sign-off

Every audit trail gate — strategy approval, cash funding, allocation execution — requires human action. These gates cannot be automated.

Instrument selection

Final instrument selection is a human decision made by the Stock Analyst from the screened universe. Screening reduces the universe; it does not select the portfolio.

4. THE BLACK BOX QUESTION

Addressed directly — because it is the right question to ask.

> *"Is Vektor a black box?" — No. Here is the precise answer.*

Strategy record transparency

For every instrument in every strategy, a PM can inspect: which indicator was assigned, what parameters were optimised, the backtest Sharpe ratio, and whether XGBoost validation passed or filtered the signal.

Model output visibility

Confidence score and pass/fail classification stored in the strategy record. PM sees which signals passed and which were filtered.

Allocation transparency

Complete breakdown: instrument, target weight, live price, lot-rounded quantity, target value, efficiency figure. Nothing summarised.

Audit trail completeness

Every action logged with timestamp and user or service identity. Complete chain from model inference to trade-ready position is retrievable.

5. HUMAN APPROVAL GATES

Every point in the workflow where human action is required before capital can move.

GATE	ROLE	WHAT IS REQUIRED
Strategy approval	Portfolio Manager	PM reviews all instruments, weights, indicators, and XGBoost validation results. Explicit approval action required. No strategy can be assigned to a client before this record exists.
Client assignment	Portfolio Manager	PM creates client profile and assigns the approved strategy. Separate explicit action from strategy approval.
Allocation execution	Portfolio Manager	PM reviews full breakdown (symbols, quantities, prices, efficiency) and confirms. Cannot proceed without explicit PM confirmation.
Cash funding	Operations	Operations records wire transfer with currency, amount, and reference. Operations cannot approve strategies or initiate allocations — structurally separated from the investment decision chain.
Market order submission (post-approval)	System	SQS delivers orders to IBKR Gateway only after strategy approval and cash funding are complete. Executes only positions a human PM has already approved in full.

6. MODEL GOVERNANCE

How models are trained, deployed, monitored, and updated.

Model training

XGBoost trained in SageMaker using three years of daily price history and historical signal outcomes. Training is triggered explicitly. Each run produces a versioned model artefact stored in S3.

Model deployment

Deployment to the inference endpoint requires an explicit promotion step. New version does not automatically replace the live model. Previous version available for immediate rollback.

Drift detection

Automated observability monitoring is on the near-term roadmap. Currently, model performance is reviewed when signal validation results appear anomalous. Formal drift detection will be implemented as part of the production observability work.

Anomalous output response

If the endpoint produces output outside the expected confidence range, the signal is flagged rather than passed or filtered. The PM sees the flag and decides. The model cannot produce an outcome that bypasses the PM review gate.

Rollback

Rollback to a previous version is a single SageMaker endpoint update. Previous artefact remains in S3. All versions retained.

7. REGULATORY ALIGNMENT — THREE JURISDICTIONS

How the Vektor AI governance framework relates to published regulatory guidance across Singapore, Hong Kong, and the United Kingdom.

> Three financial regulators — MAS (Singapore), SFC/HKMA (Hong Kong), and FCA/PRA (United Kingdom) — have independently published guidance on responsible AI use in financial services. Despite different regulatory traditions and market contexts, all three converge on the same four requirements: (1) explainability of AI outputs, (2) human oversight at decision points, (3) clear accountability for AI-assisted decisions, and (4) the ability to audit and review AI behaviour. The Vektor governance framework was designed against these shared first principles — not against any single jurisdiction's specific rules. This makes it durable as regulation evolves and portable across markets.

VEKTOR'S ALIGNMENT WITH THE FOUR SHARED REQUIREMENTS

1. Explainability

Regulatory requirement: AI outputs must be explainable and interpretable.

Vektor alignment: The XGBoost model produces a confidence score and pass/fail for each signal — both stored in the strategy record and visible to the PM. Training data window, feature set, and threshold are documented. A compliance reviewer can trace any signal back to the model version, training data, and confidence score that produced it.

2. Human oversight

Regulatory requirement: Human oversight must be maintained at AI-influenced decision points.

Vektor alignment: Human approval gates exist at every capital-at-risk stage. Strategy approval, allocation execution, and cash funding are explicit human actions. The XGBoost output is an input to the PM review, not a substitute for it.

3. Accountability

Regulatory requirement: Clear responsibility must exist for AI-assisted decisions.

Vektor alignment: Every AI output is attributable to a specific model version, training dataset, and inference request. Every investment decision that follows is attributable to a specific human — PM identity, timestamp, action — in the audit log.

4. Auditability

Regulatory requirement: AI behaviour must be auditable and reviewable.

Vektor alignment: Every model call, signal validation result, strategy approval, and allocation is logged with timestamp and identity. The complete chain from inference to trade-ready position is retrievable at any point.

JURISDICTIONAL REFERENCES

Singapore — MAS

— MAS FEAT Principles (Fairness, Ethics, Accountability, Transparency) — Veritas initiative, 2019 onwards

— MAS Guidelines on the Use of Artificial Intelligence and Data Analytics in Financial Services

The MAS FEAT framework is the most developed AI governance framework in the ASEAN region. Its four principles map directly to Vektor's governance design. MAS's Veritas initiative provides implementation guidance for financial services firms seeking to operationalise FEAT.

Hong Kong — SFC / HKMA

— SFC Circular on Algorithmic Trading — March 2018 (reinforced in subsequent guidance)

— HKMA Supervisory Policy Manual TM-G-1 — Technology Risk Management: explainability and human oversight of automated systems

The SFC's algorithmic trading circular requires that automated trading systems have appropriate governance, testing, and human oversight before deployment. HKMA TM-G-1 addresses technology risk management and specifically requires explainability and human oversight of automated systems — directly relevant to XGBoost signal validation. Vektor's documented model governance and human approval gates are consistent with both frameworks.

United Kingdom — FCA / PRA

— PRA Supervisory Statement SS1/23 — Model Risk Management Principles (supervisory expectation, not advisory guidance)

— FCA DP5/22 — Artificial Intelligence and Machine Learning (discussion paper indicating regulatory direction)

PRA SS1/23 is the most directly applicable document — it is supervisory expectation, not advisory guidance, and requires model explainability, governance, and the ability to intervene in automated processes. The FCA's DP5/22 signals the direction of future binding rules. Vektor's framework — documented model versions, explicit promotion steps, rollback capability, and human approval at every investment gate — is consistent with SS1/23 expectations.

> The regulatory landscape for AI in financial services is evolving rapidly across all three jurisdictions. The Vektor governance framework is designed to be durable — anchored to the shared first principles that regulators are converging on, not tied to any single jurisdiction's current rules. Regulatory standing note: Vektor is a portfolio construction platform. Regulatory status and applicable permissions vary by jurisdiction and use case. Full details are available to Proof Partners on request.

8. AI CAPABILITY ROADMAP

Future ML applications under consideration — each evaluated against the same governing principle.

Each future capability will be evaluated against the same question: does it perform a computation task better than human analysis, without displacing a human decision that carries accountability?

Formal walk-forward validation

Out-of-sample validation of XGBoost model against live performance data. Directly addresses the overfitting risk acknowledged in WP-04.

Automated drift detection

CloudWatch-based monitoring of XGBoost inference quality, triggering alerts when patterns diverge from historical norms. Output: alert to a human reviewer — not automatic model retraining.

Enhanced signal universe

Evaluation of additional technical indicators or alternative data sources. Any new signal type goes through the same grid search and XGBoost validation before inclusion in production strategies.

Rebalancing trigger detection

ML-assisted identification of when positions have drifted materially from target weights. Output: a rebalancing recommendation to the PM — not an automatic trade.

9. CONCLUSION

The governing framework does not change as the technology develops — the technology is evaluated against the framework.

AI is used in Vektor where it outperforms human analysis on a well-defined, measurable task. It is not used where the decision carries accountability and requires contextual judgement. This boundary is structural — enforced by the platform's workflow architecture, not by a policy that could be overridden.

The governance framework described in this paper aligns with the shared principles that financial regulators in Singapore, Hong Kong, and the United Kingdom have independently arrived at. That convergence is not coincidence — it reflects a shared view that AI in financial services must augment human judgement, not replace it. Vektor was built on that view from the start.

For technical infrastructure supporting ML deployment, see WP-07. For compliance and audit trail architecture, see WP-06. For signal optimisation methodology, see WP-02. All documents at investpuppy.com.